

Spotter Machine Learning Engineer Assessment

Freight Rate Prediction — Technical Submission Report

Objective. Predict freight load rates for 12,000 future loads while demonstrating leakage-safe validation, robust data handling, feature engineering, model selection, and reproducible delivery.

Executive Summary

The development set contains 48,000 labeled loads from January through October 2025. The assessment validation set contains 12,000 future loads from November and December. Because the prediction horizon is strictly later than the labeled period, I used chronological validation rather than a random split. The final pipeline predicts log rate-per-mile with robust LightGBM regressors and blends a market-aware model with a market-agnostic model.

On the primary January–August → September–October holdout, the selected ensemble achieved **MAE \$104.46**, **RMSE \$633.92**, **MAPE 4.42%**, and **R² 0.827**. A recent October-only stress test achieved MAE of **\$96.68**.

1. Data Understanding and Quality

Item	Finding	Treatment
Development data	48,000 rows; Jan–Oct 2025	Chronological modeling
Future validation	12,000 rows; Nov–Dec 2025	Predicted without labels
Negative weights	292 rows	Absolute value + missingness flag
Missing weights	300 rows	Development median imputation + flag
Missing market index	374 rows	Development median imputation + flag
Extreme rate/mile	772 rows (1.61%) outside 0.30–3.00	Robust log target + L1 loss; no row deletion

The target is strongly distance-driven: the development-set Pearson correlation between distance and posted rate is approximately 0.909. There are 64 pickup cities, 64 delivery cities, and 4,014 unique pickup/delivery routes. The target median is \$2,030.76 and the mean is \$2,373.98, indicating meaningful right-tail behavior.

2. Target Formulation and Feature Engineering

Rate-per-mile target. I model `posted_rate / distance` rather than raw dollars. The final prediction is reconstructed as $\exp(\text{predicted log rate-per-mile}) \times \text{distance}$. This reduces the burden on the model to relearn the near-linear distance effect and improves stability across short and long lanes.

Feature groups. The pipeline includes route and route-equipment categorical structure; cleaned weight and weight-per-mile; calendar variables; seasonal sine/cosine terms; pickup/delivery coordinates; haversine distance; coordinate deltas and direction; market index; quote signal; and distance $\times$ market/quote interactions.

For the fixed December chart, market index and quote signal are intentionally excluded because they are absent from the provided seven-column chart schema. City-level coordinates are recovered from medians learned from the development set, avoiding invented geographic values.

3. Validation Strategy

A random split was rejected because it would allow later observations to influence training while estimating performance on an earlier period. The chronological design matches the actual assessment deployment scenario.

Backtest	Train	Holdout	MAE	RMSE	MAPE	R ²
Primary	Jan–Aug	Sep–Oct	\$104.46	\$633.92	4.42%	0.827
Recent stress	Jan–Sep	Oct	\$96.68	\$645.51	4.48%	0.822

4. Model Selection and Final Architecture

The final architecture uses two robust LightGBM regressors. Both use a shallow-tree configuration, learning rate 0.03, L1 regression objective, and a log rate-per-mile target clipped only during target construction to reduce the influence of extreme multiplicative noise.

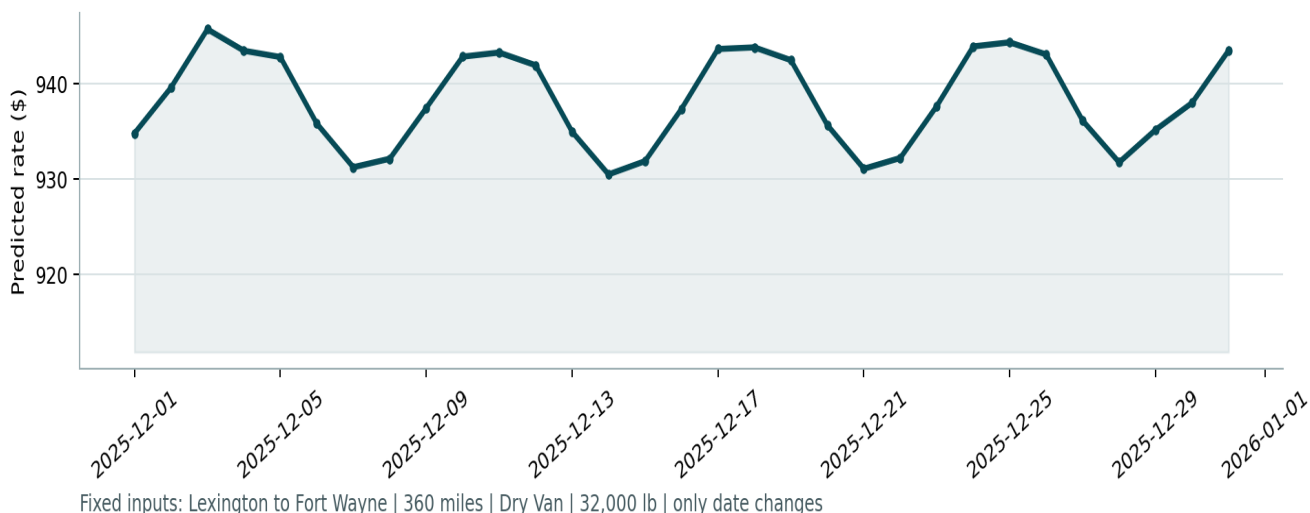
Model A — market-aware: 700 estimators, 7 leaves, random seed 3. **Model B — market-agnostic:** 450 estimators, 7 leaves, random seed 7. Final validation predictions use a 70/30 blend of A/B. The second model is a deliberate robustness component for missing or unstable market variables.

The most influential model features include route, quote signal, weight-per-mile, equipment, day-of-year, route-equipment interaction, pickup, delivery, weight, and market index. This is consistent with the freight-pricing structure visible in the data.

5. December Scenario

The assessment supplies 31 fixed December rows for Lexington → Fort Wayne, 360 miles, Dry Van, and 32,000 lb, with only the date changing. Because market and quote fields are absent from this scenario, the December forecast uses a chart-safe model that consumes only fields actually available in the scenario. The resulting curve remains positive and exhibits calendar-driven variation rather than a fabricated market forecast.

Candidate: December 2025 Predicted Load Rate



The comparable development-data lane history contains 10 loads matching the same route/equipment and approximately the same distance/weight band, with a median posted rate of about \$809.43. The December chart is therefore also checked against the observed lane scale as a sanity constraint.

6. Reproducibility and Submission Contract

The repository includes the training pipeline, feature engineering module, dependencies, scorer, prediction outputs, report, and Loom script. The official output contract was run successfully: 12,000 validation predictions and 31 December predictions were accepted by the scorer, and the December chart was generated.

The validation predictions contain exactly the required columns `load_id`, `predicted_rate`. All 12,000 load IDs match the supplied template, predictions are finite and positive, and the December file preserves its required seven-column schema and all 31 dates.

7. Limitations

The true `posted_rate` values for the final 12,000 validation loads are hidden by the assessment. Therefore, the internal holdout metrics are model-selection evidence rather than the final Spotter score. The final model is intentionally trained on all 48,000 labeled development rows after model selection.

Conclusion

The solution emphasizes leakage-safe temporal validation, robust target formulation, explicit data-quality treatment, route-aware nonlinear modeling, and strict output validation. It is designed to be reproducible and easy for a reviewer to audit from raw data through final prediction files.